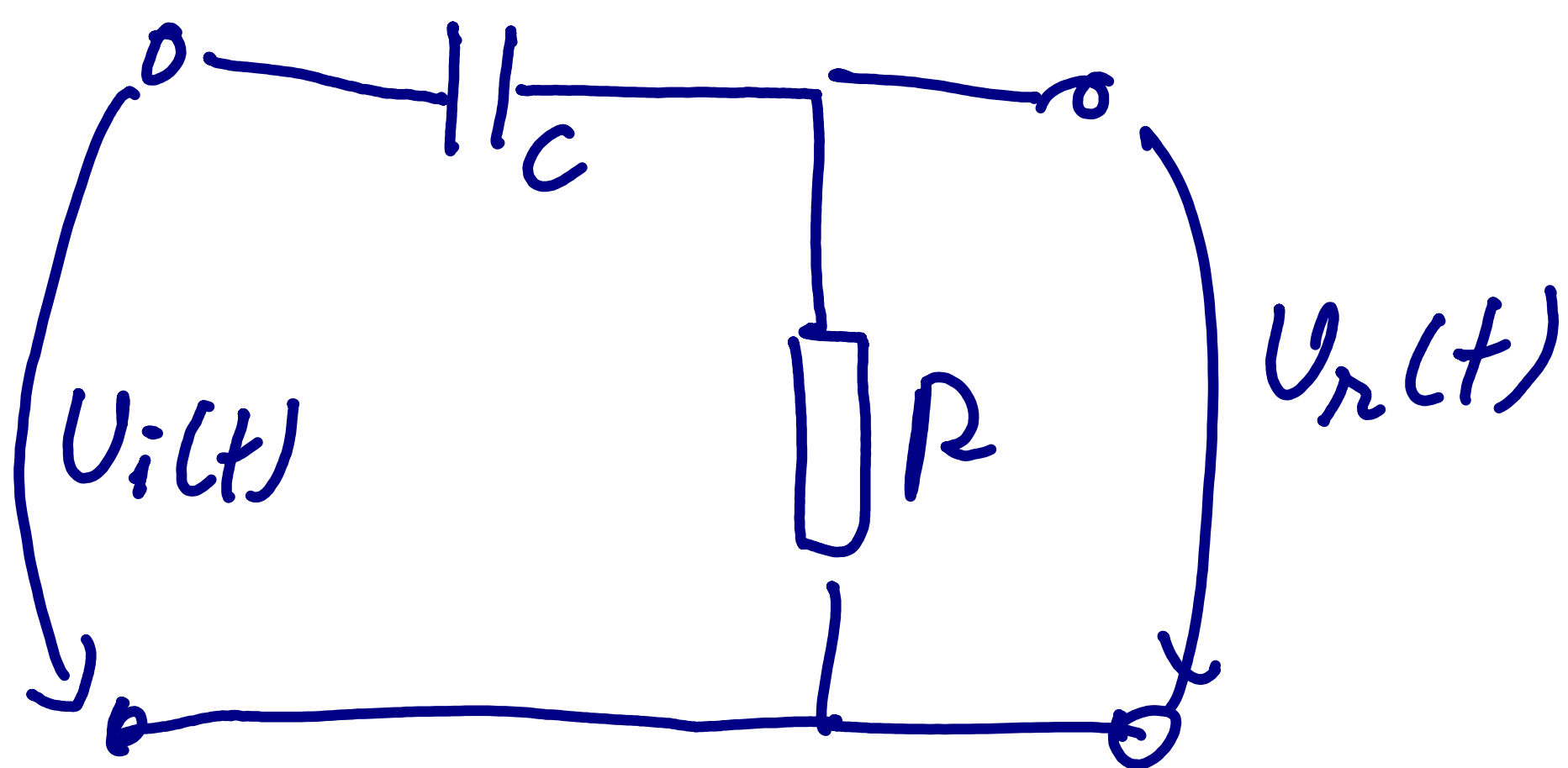


27.10.2025



$$u(t) = u_i(t)$$

$$y(t) = u_R(t)$$

$$u_i(t) = u_c(t) + u_R(t)$$

$$u(t) = u_c(t) + y(t)$$

$$u_c(t) = \frac{1}{C} \int_0^t i(\tau) d\tau \Rightarrow \frac{du_c(t)}{dt} = \frac{1}{C} i(t) \Rightarrow i(t) = C \frac{du_c(t)}{dt}$$

$$u_R(t) = R \cdot i(t)$$

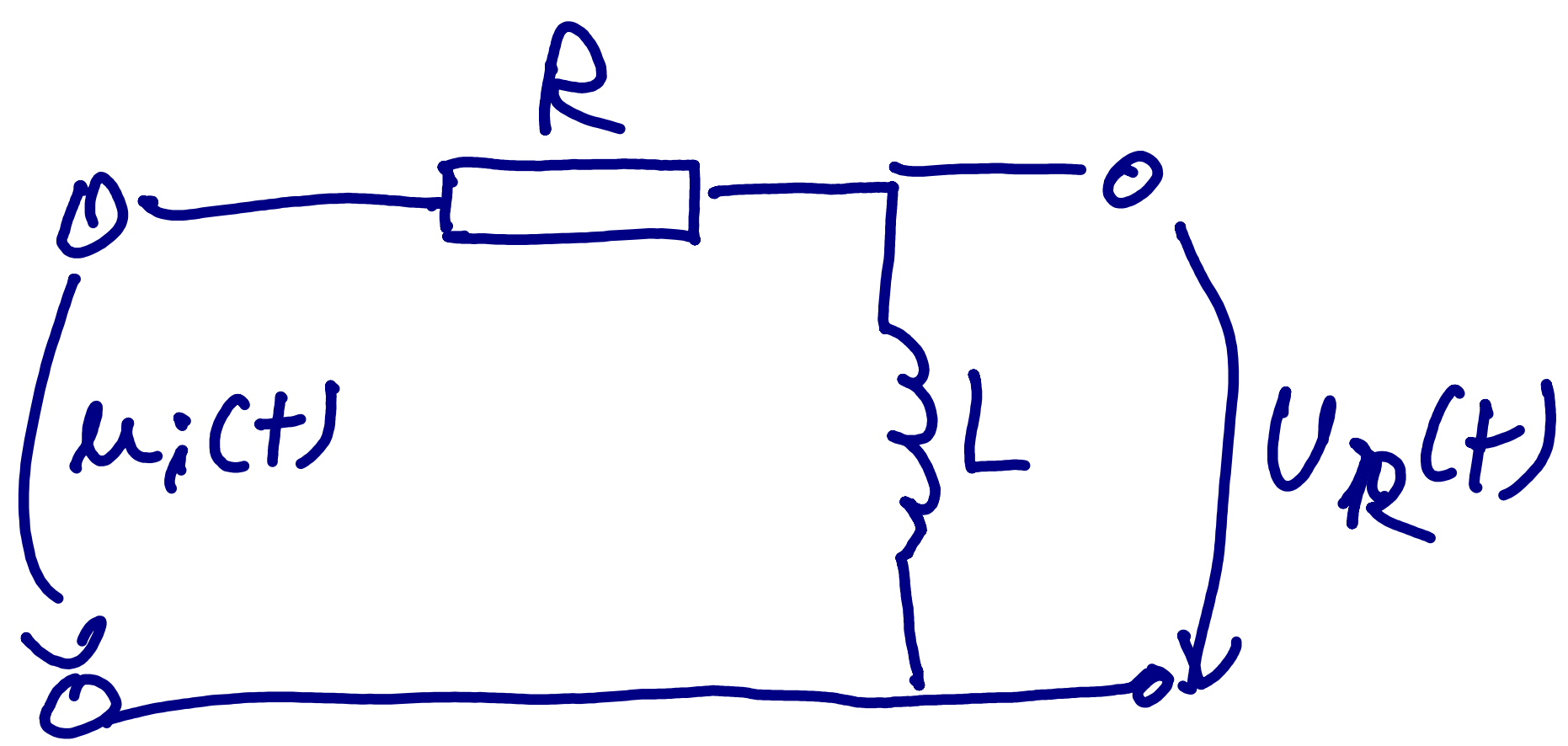
$$u(t) = \frac{1}{C} \int_0^t i(\tau) d\tau = y(t)$$

$$u_c(t) = u(t) - y(t)$$

$$\dot{i}(t) = C \frac{d(u(t) - y(t))}{dt} = C \left(\frac{du(t)}{dt} - \frac{dy(t)}{dt} \right)$$

$$i(t) = \frac{u_R(t)}{R} \Rightarrow y(t) = RC \left(\frac{du(t)}{dt} - \frac{dy(t)}{dt} \right)$$

$$y(t) + \frac{dy(t)}{dt} RC = RC \frac{du(t)}{dt}$$



$$u(t) = u_i(t)$$

$$y(t) = u_L(t)$$

$$u_i(t) = u_R(t) + u_L(t)$$

$$u(t) = u_R(t) + y(t)$$

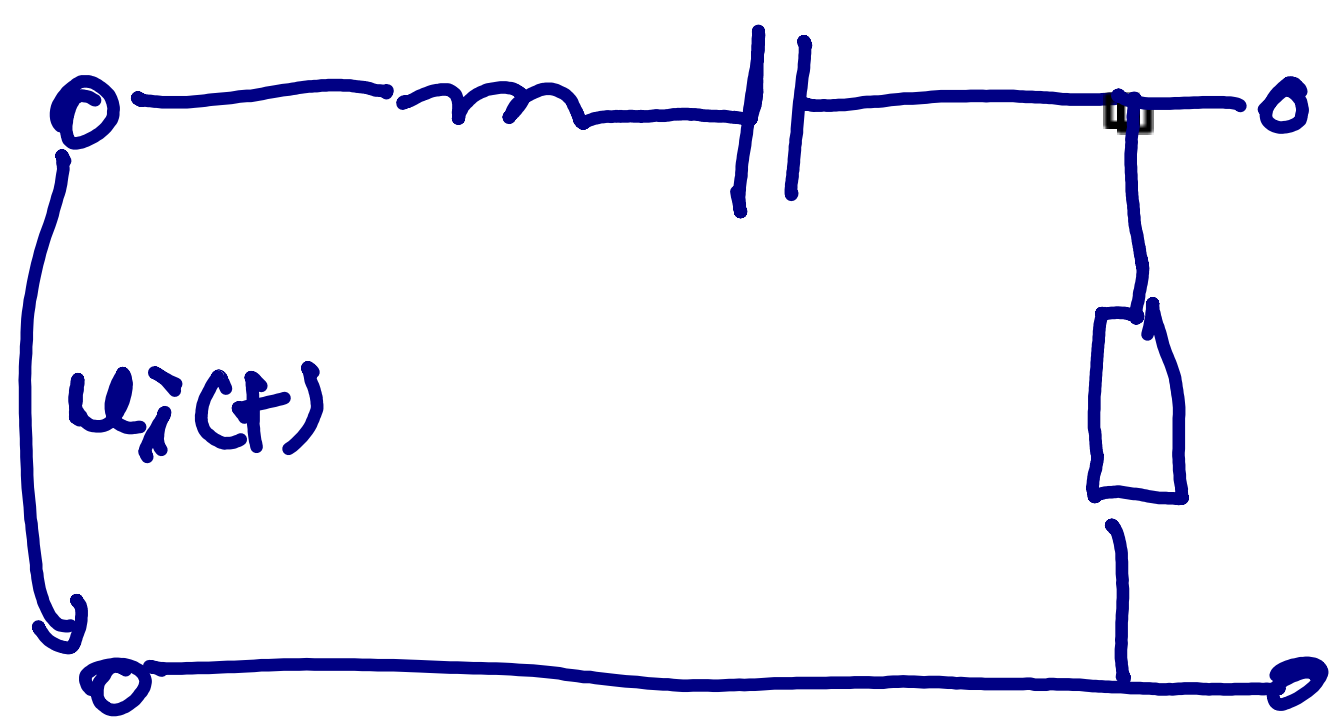
$$u_R(t) = R \cdot i(t)$$

$$u_L(t) = L \frac{di(t)}{dt} \Rightarrow i(t)L = \int_0^t u_L(\tau) d\tau \Rightarrow i(t) = \frac{1}{L} \int_0^t y(\tau) d\tau$$

$$u(t) = R \cdot i(t) + y(t)$$

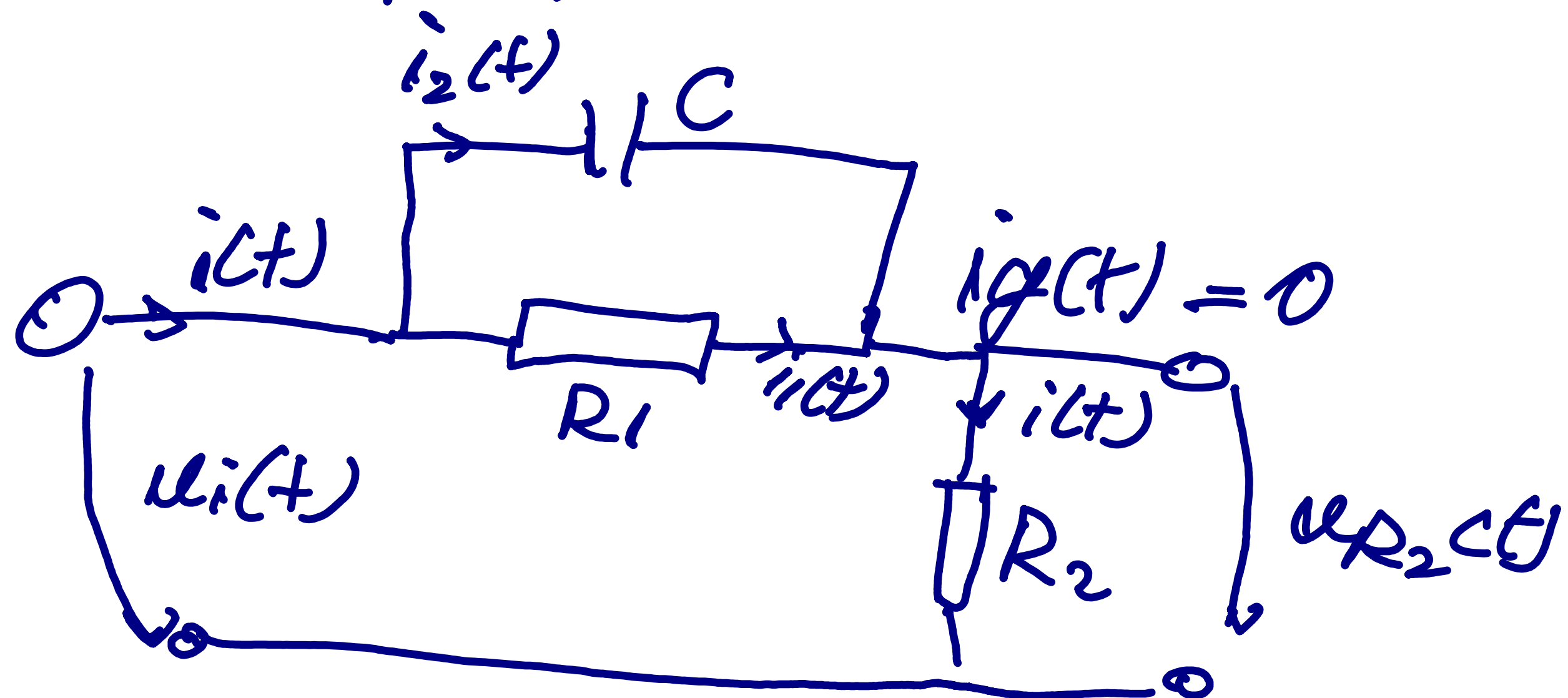
$$u(t) = \frac{R}{L} \int_0^t y(\tau) d\tau + y(t) \quad \left| \frac{d}{dt} \right.$$

$$\frac{du(t)}{dt} = \frac{R}{L} y(t) + \frac{dy(t)}{dt}$$



$$u(t) = u_i(t)$$

$$u(t) = u_R(t)$$



$$i(t) = i_1(t) + i_2(t)$$

$$u_i(t) = u_{R_1}(t)$$

